

OSN2526: Assignment 2: Page tables

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1 Inleiding

This report describes an analysis of different virtual memory implementations. In Section 2 a comparative analysis is performed between a simple single level MMU and a RISC-V Sv48 MMU. Both MMU are simulated within the provided framework given for the assignment.

2 MMU Comparison

This section performs a comparative analysis of a simple single level MMU provided for the assignment and a RISC-V Sv48 MMU implementation.

Table 1 shows the results for a single process and Table 2 shows the results for multiple processes.

Tabel 1: Single process results of RISC-V Sv48 and Simple MMU Implementations for trace: gcc.txt.gz.

Metric	RISC-V (Sv48)	Simple (Single-Level)	Difference
Page size	4 KB	64 MB	-
Page table allocated	36 KiB	64 MiB	67 MB
Handled page faults	6	4	2
Max. # allocated physical pages	15	5	10

Tabel 2: Multiprocess results of RISC-V Sv48 vs. Simple MMU (gcc.txt.gz and ls.txt.gz)

Metric	RISC-V (Sv48)	Simple (Single-Level)	Difference
Page size	4 KB	64 MB	-
Page table allocated	68 KiB	128 MiB	134 MB
Handled page faults	11	7	4
Max. # allocated physical pages	28	8	20

These results nicely capture the impact of using a RADIX tree to manage the virtual memory translation. For a single level table each store of a physical memory is accompanied by a 64 MB page size allocation. When we have two physical addresses which are further apart than the size of the page the amount of bytes allocated easily duplicates. For the multilevel RISC-V Sv48, this issue is not as severe since the pages are only of size 4 KB. When a fragmented access occurs the relative allocation overhead is much

less. This can be seen in the results, the Sv48 implementation allocates 36 KiB for a single process compared to 64MiB for the single level implementation. Another memory efficiency is captured by the multiprocess results, each process only has to allocate per 4KB chunks for the Sv48 implementation.

This does go at the cost of some performance. The implementation of the Sv48 is more complex, it needs to go through a couple of stages/levels to retrieve the physical address. Also, since the page size is lower it covers less physical address space, thus increase the amount of page faults.

3 Concluse